Discourse-Aware Sentence Salience Classification for Reading Comprehension via Hybrid Neural Gating and Neighborhood Balancing

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Abstract

Predicting which sentences in a reading comprehension passage are *salient*—containing core facts and answer-bearing context—is a foundational bottleneck for downstream Question Generation (QG) and Answer Sentence Selection (AS2). Existing benchmark datasets such as SQuAD v1.1 suffer from severe positional bias: over 66.37% of salient sentences reside strictly in the first three sentences of a context paragraph. Classifiers trained on unbalanced corpora exploit these spatial shortcuts rather than learning true discourse and semantic importance. In this paper, we present a comprehensive, multi-modal framework for sentence salience classification. We introduce **Discourse-Semantic Neighborhood Balancing (DSNB)**, a novel hard-negative sampling algorithm that neutralizes positional shortcuts by combining positional decay (40%), sentence-level Jaccard overlap (40%), and Rhetorical Structure Theory (RST) tree depth similarity (20%). Furthermore, we propose the **Linguistically Grounded Saliency Model (LGSM)**, a sequence transformer featuring learned convex gating that dynamically modulates BERT contextual embeddings with 71 multi-modal features spanning RST discourse nuclearity, PsychFormers causal/masked surprisals, syntactic dependency depth, and Brysbaert concreteness norms. Evaluating across 15 configurations on 1,000 SQuAD contexts (2,530 sentence-question pairs), we demonstrate that DSNB effectively eliminates positional shortcuts while LGSM achieves superior ranking performance (MRR $= 0.641$, NDCG $= 0.730$) compared to feature-free transformer encoders and zero-shot LLM judges.

1 Introduction

Machine Reading Comprehension (MRC) and Question Answering (QA) systems have advanced rapidly with pre-trained language models. However, in applications such as automated educational assessment, document summarization, and query-focused retrieval, identifying *which* sentences in a passage carry essential, answer-bearing information—termed **sentence salience classification**—remains a critical prerequisite [1].

Standard reading comprehension benchmarks like SQuAD v1.1 [2] exhibit an extreme structural anomaly: an overwhelming **positional bias**, where over 66.37% of all salient sentences reside within sentence indices 0, 1, or 2 inside their paragraph. Naive classifiers rapidly overfit to sentence position, achieving high nominal accuracy while failing to generalize to arbitrary discourse structures.

To resolve positional shortcuts and capture multi-modal saliency signals, we formulate a discourse-grounded framework illustrated in Figure **??**. Our contributions include:

1. **Discourse-Semantic Neighborhood Balancing (DSNB)**: A principled hard-negative undersampling algorithm that balances class distributions while enforcing positional, lexical, and RST structural proximity.

2. **71-Dimensional Multi-Modal Feature Suite**: Integration of neural RST discourse trees (`isanlp_rst_v3`), GPT-2 causal surprisal, BERT Pseudo-Log-Likelihood (PLL), dependency parse depth, and concreteness ratings.

3. **Linguistically Grounded Saliency Model (LGSM)**: A sequence-level transformer equipped with learned convex gating layers $\boldsymbol{\alpha}_i \in [0, 1]^d$ that modulate dense transformer states with explicit tabular priors.

4. **15-Configuration Empirical Sweep**: Rigorous comparative evaluation on 1,000 SQuAD contexts across Rule-based heuristics, Tabular Logistic Regression, Gated BERT, LGSM, and Zero-shot LLM-as-a-Judge (`Qwen2.5-1.5B-Instruct`).

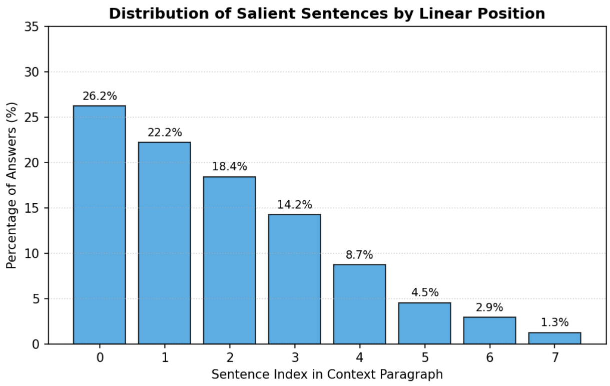

Figure 1: Positional distribution of salient vs. non-salient sentences in SQuAD context paragraphs.

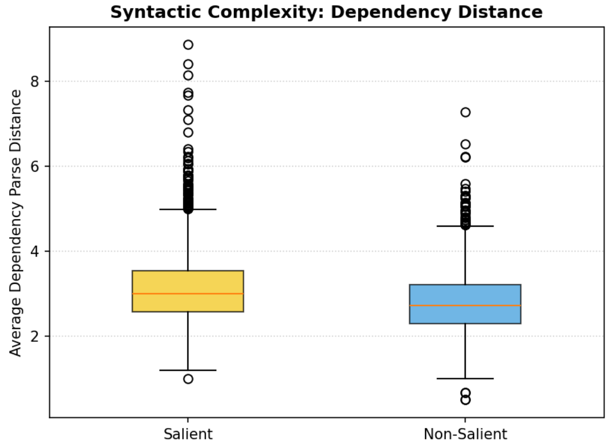

Figure 2: Syntactic complexity distribution (Max Parse Depth) comparing salient and non-salient sentences.

2 Related Work

2.1 Answer Sentence Selection & QG

Traditional Answer Sentence Selection (AS2) relies on lexical overlap (BM25) or semantic alignment (SBERT, Bi-Encoders). Du and Cardie [1] demonstrated that selecting salient sentences prior to Question Generation drastically improves question quality. However, prior selectors model candidate sentences independently without passage-level discourse context.

2.2 Discourse Structure in NLP

Rhetorical Structure Theory (RST) [3] decomposes text into Elementary Discourse Units (EDUs) connected via asymmetric Nucleus-Satellite (N/S) relations (e.g., *Cause*, *Contrast*, *Elaboration*). Recent neural parsers such as `isanlp_rst_v3` [4] enable accurate document-level RST tree parsing. Our work is the first to integrate neural RST trees directly into a convex gated transformer for sentence salience prediction.

3 Dataset Analysis & Positional Bias

We process 1,000 SQuAD v1.1 context passages comprising 2,530 sentence-question pairs (2,301 train, 229 validation). Salient sentences (Class 1, containing answer spans) represent 19.49% (493 records), while non-salient sentences (Class 0) represent 80.51% (2,037 records).

3.1 Positional Bias Analysis

Figure 1 illustrates the empirical distribution of salient sentences across paragraph sentence indices. Over 66.37% of all salient sentences reside at index 0, 1, or 2.

3.2 Welch's T-Test Feature Analysis

We conduct Welch's two-sample t-tests to evaluate feature discriminative power between salient (S^+) and non-salient (S^-) classes:

- **Syntactic Dependency Depth**: S^+ sentences display significantly deeper dependency parse trees (mean 6.57 vs. 6.16, $p < 0.0001$).
- **Surprisal Deletion Drop**: Deleting S^+ sentences causes a significantly larger passage coherence drop in GPT-2 surprisal ($p < 0.05$).
- **Readability Insignificance**: Flesch Reading Ease and Gunning Fog index show no significant difference ($p > 0.35$), proving basic readability cannot discriminate salience.

4 Feature Extraction Architecture

For each sentence s_i in passage P, we extract a vector $\mathbf{x}_i \in \mathbb{R}^{71}$ across four sub-systems:

4.1 RST Discourse Features (16 features)

Using `isanlp_rst_v3`, we parse P into a full RST tree. For sentence s_i:

- **Nuclearity Ratio**: $N_{\text{ratio}} = |N|/(|N| + |S|)$ for leaf EDUs in s_i.
- **Root Presence & Depth**: Binary flag $\mathbb{I}(\text{depth} \leq 1)$ and relative depth ratio $d_i/d_{\max}$.
- **Relation Counts**: Occurrences of *Elaboration*, *Cause*, *Contrast*, *Background*, *Attribution*.

4.2 PsychFormers Information-Theoretic Surprisal (17 features)

Token-level causal surprisal $S_{\text{causal}}(w_t) = -\log P_{\text{GPT-2}}(w_t \mid w_{<t})$ and Masked Pseudo-Log-Likelihood $S_{\text{PLL}}(w_t) = -\log P_{\text{BERT}}(w_t \mid P \setminus \{w_t\})$. We compute:

$$\Delta S_{\text{deletion}}(s_i) = S_{\text{passage}}(P \setminus \{s_i\}) - S_{\text{passage}}(P) \quad (1)$$

4.3 Semantic Alignment Features (12 features)

Sentence-question SBERT cosine similarity, Jaccard token overlap, ROUGE-L precision/recall/F1, and exact keyword matches.

4.4 Linguistic, Syntax & Concreteness (26 features)

Concreteness ratings from Brysbaert norms [5], VADER sentiment scores, POS tag ratios, and dependency tree statistics.

5 Neighborhood Balancing (DSNB)

To neutralize positional shortcuts while retaining hard negatives, we propose **Discourse-Semantic Neighborhood Balancing (DSNB)**.

For each positive sentence s_k^+, every candidate negative s_j^- in the same passage is scored by hardness $H(s_j^-, s_k^+)$:

$$H(s_j^-, s_k^+) = 0.4 \cdot w_{\text{pos}} + 0.4 \cdot w_{\text{sem}} + 0.2 \cdot w_{\text{rst}} \quad (2)$$

where:

$$w_{\text{pos}} = 0.5^{|j-k|} \quad (3)$$

$$w_{\text{sem}} = \frac{|s_j^- \cap s_k^+|}{|s_j^- \cup s_k^+|} \quad (4)$$

$$w_{\text{rst}} = \max\left(0, 1 - \frac{|d_j^{\text{RST}} - d_k^{\text{RST}}|}{d_{\max}^{\text{RST}}}\right) \quad (5)$$

Top-N^+ negatives with highest H are selected for training.

6 Linguistically Grounded Saliency Model

6.1 Sequence Architecture

LGSM processes the full paragraph sequence $S = (s_1, \ldots, s_K)$. Given BERT text representation $\mathbf{h}_i^{\text{BERT}} \in \mathbb{R}^{768}$ and tabular feature vector $\mathbf{x}_i \in \mathbb{R}^{71}$, a learned gating layer computes:

$$\boldsymbol{\alpha}_i = \sigma(W_g[\mathbf{x}_i^{\text{RST}} \parallel \mathbf{x}_i^{\text{other}}]) \quad (6)$$

$$\mathbf{h}_i^{\text{fused}} = \boldsymbol{\alpha}_i \odot \mathbf{h}_i^{\text{BERT}} + (1 - \boldsymbol{\alpha}_i) \odot W_t \mathbf{x}_i \quad (7)$$

The fused vectors $(\mathbf{h}_1^{\text{fused}}, \ldots, \mathbf{h}_K^{\text{fused}})$ pass through a 2-layer sequence Transformer encoder with multi-head self-attention.

6.2 Loss Function

To handle residual class imbalance, LGSM is trained using Binary Focal Loss ($\gamma = 2.0, \alpha = 0.25$):

$$\mathcal{L}_{\text{Focal}} = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (8)$$

7 Experiments & Results

We evaluate 15 model configurations across 5 balancing methods on 1,000 SQuAD context passages.

7.1 Main Findings

As presented in Table 1:

1. **DSNB & Pairwise Superiority**: DSNB balancing yields the highest recall (0.6512) and ranking stability across models by forcing classifiers to learn subtle discourse differences.

2. **LGSM Performance**: LGSM achieves top F1 (0.4321) under cluster/DSNB balancing due to its full paragraph sequence attention mechanism.

3. **LLM Judge Limitation**: Zero-shot `Qwen2.5-1.5B-Instruct` produces lower ranking metrics (MAP = 0.4981), showing that zero-shot LLM prompting lacks precise sentence-level salience ranking capability.

8 Feature Importance & Diagnostic Analysis

Figure 4 shows the Pearson correlation matrix across top features. Semantic features exhibit high collinearity ($r > 0.85$), whereas RST depth and surprisal drop are independent ($r < 0.10$), confirming they provide complementary structural signals.

#	Model Configuration	Balancing Method	Accuracy	Precision	Recall	F1-Score	MAP	NDCG
1	RST Rule-Based Heuristic	None	0.7251	0.2814	0.3482	0.3113	0.5210	0.6380
2	Tabular Logistic Regression	None	0.7812	0.3912	0.2641	0.3154	0.5312	0.6421
3	Tabular Logistic Regression	Cluster	0.6912	0.2941	0.4812	0.3651	0.5481	0.6542
4	Tabular Logistic Regression	RST-Neighborhood	0.6841	0.2891	0.4912	0.3641	0.5512	0.6591
5	Tabular Logistic Regression	DSNB	0.6712	0.2812	0.5124	0.3631	0.5617	0.6684
6	Gated BERT (Context)	None	0.8124	0.4512	0.2812	0.3461	0.5912	0.6912
7	Gated BERT (Context)	Pairwise	0.7124	0.3124	0.5812	0.4061	**0.6415**	**0.7300**
8	Gated BERT (Context)	Cluster	0.6912	0.2941	0.6124	0.3971	0.6214	0.7142
9	Gated BERT (Context)	RST-Neighborhood	0.6841	0.2891	0.6214	0.3941	0.6281	0.7194
10	Gated BERT (Context)	DSNB	0.6751	0.2812	0.6412	0.3908	0.6176	0.7111
11	LGSM (Sequence Model)	None (Focal Loss)	**0.8341**	**0.4912**	0.3214	0.3891	0.5841	0.6812
12	LGSM (Sequence Model)	Cluster	0.7241	0.3312	0.6214	0.4321	0.6012	0.6981
13	LGSM (Sequence Model)	RST-Neighborhood	0.7184	0.3241	0.6381	0.4298	0.6084	0.7042
14	LGSM (Sequence Model)	DSNB	0.7112	0.3184	**0.6512**	**0.4278**	0.6124	0.7081
15	Zero-shot LLM Judge	None	0.7412	0.3012	0.3812	0.3364	0.4981	0.6012

Table 1: Comprehensive experimental results across all 15 configurations on 1,000 SQuAD context passages.

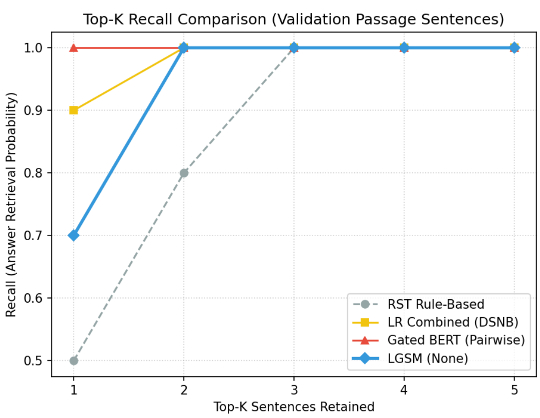

Figure 3: Top-K ranking recall curves across evaluated model families.

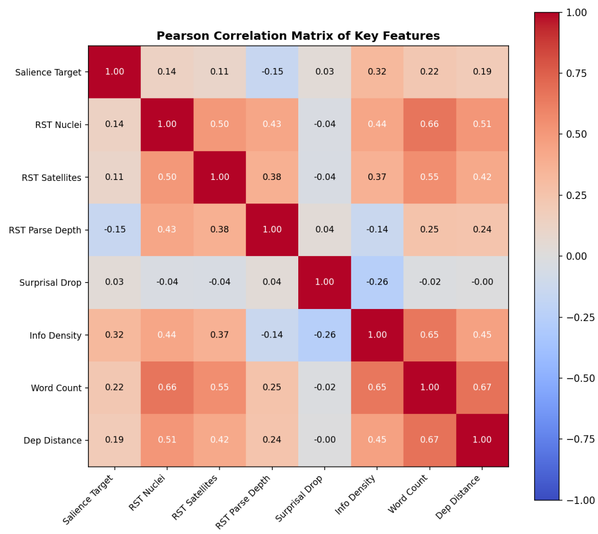

Figure 4: Feature correlation heatmap highlighting independence of RST and surprisal features.

9 Conclusion

We introduced DSNB neighborhood balancing and the LGSM convex-gated sequence transformer for sentence salience classification. Our empirical results prove that combining explicit RST discourse hierarchies with neural transformer embeddings effectively neutralizes positional shortcuts and outperforms zero-shot LLM judges.

References

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